# Chapter 5 — Solids and Fluid Dynamics

Full textbook exercise · PECTAA Physics 11 · No ALP filtering

## Q.1 — Multiple Choice Questions (with correct option and reason)

### 5.1. The region of stress–strain curve which obeys Hooke's law is:

**(a)** proportional limit **(b)** elastic region
**(c)** plastic region **(d)** yield limit

**Answer: (a) proportional limit** — Stress is proportional to strain up to the proportional limit.

### 5.2. Which of the following is more elastic?

**(a)** rubber **(b)** wood
**(c)** sponge **(d)** steel

**Answer: (d) steel** — Steel has a much larger Young's modulus.

### 5.3. Which of the following is a polymer solid?

**(a)** wool **(b)** glass
**(c)** sodium chloride **(d)** copper

**Answer: (a) wool** — Wool is a natural polymeric solid.

### 5.4. Decrease of pressure with increase of speed in a horizontal pipe is:

**(a)** Torricelli's effect **(b)** Bernoulli's effect
**(c)** Venturi's effect **(d)** Doppler's effect

**Answer: (b) Bernoulli's effect** — It is Bernoulli's principle.

### 5.5. The pressure will be low when speed of a fluid is:

**(a)** zero **(b)** high
**(c)** low **(d)** constant

**Answer: (b) high** — $P + \frac{1}{2}\rho v^2$ is constant in horizontal flow.

### 5.6. As per law of fluid friction for steady streamline flow, the friction:

**(a)** varies proportionally to velocity of fluid **(b)** varies inversely with pressure
**(c)** does not depend on pressure **(d)** first increases then decreases

**Answer: (a) varies proportionally to velocity of fluid** — Viscous drag $F = 6\pi\eta rv$ is proportional to speed.

### 5.7. A stone weighs less in water than in air because of:

**(a)** reduction of mass **(b)** increase of density
**(c)** buoyant force acting upwards **(d)** gravitational force acting upward

**Answer: (c) buoyant force acting upwards** — Upthrust reduces apparent weight.

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### 5.8. The principle of floatation is a direct application of:

**(a)** Pascal's law
**(b)** Bernoulli's principle
**(c)** Archimedes' principle
**(d)** Newton's third law

**Answer: (c) Archimedes' principle** — Floating depends on upthrust equal to weight.

### 5.9. An ideal flow of any fluid must satisfy:

**(a)** Pascal's law
**(b)** Bernoulli's equation
**(c)** continuity equation only
**(d)** both Bernoulli's equation and continuity equation

**Answer: (d) both Bernoulli's equation and continuity equation** — Ideal flow obeys mass and energy conservation.

### 5.10. The lift force on an aeroplane wing is primarily due to:

**(a)** viscosity of air
**(b)** density of air
**(c)** pressure difference above and below the wing
**(d)** gravitational force

**Answer: (c) pressure difference above and below the wing** — Faster air above gives lower pressure.

### 5.11. A Venturi mask delivering known oxygen concentration operates on:

**(a)** Newton's third law
**(b)** Archimedes' principle
**(c)** Pascal's law
**(d)** Bernoulli's principle

**Answer: (d) Bernoulli's principle** — Fast oxygen flow draws in air.

### 5.12. A defining characteristic of a superfluid is:

**(a)** zero viscosity
**(b)** infinite density
**(c)** zero temperature
**(d)** infinite thermal conductivity

**Answer: (a) zero viscosity** — Superfluids flow without viscosity.

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## Q.2 — Short Answer Questions

### 5.1. What is meant by (i) cohesive force (ii) viscosity?

(i) Cohesive force is attraction between molecules of the same substance. (ii) Viscosity is the internal friction of a fluid that opposes relative motion between its layers.

### 5.2. Differentiate between streamline and turbulent flow.

Streamline flow is steady and orderly, with fluid layers sliding smoothly. Turbulent flow is irregular, with eddies and mixing.

| Feature | Streamline | Turbulent |
|---|---|---|
| Pattern | Smooth layers | Eddies and mixing |
| Speed | Usually lower | Usually higher |

| Feature | Streamline | Turbulent |
|---|---|---|
| Energy loss | Small | Large |

### 5.3. How does pressure change with depth in fluids?

Pressure increases with depth according to $P = \rho gh$, because the weight of the fluid above increases.

### 5.4. How is variation in pressure related to speed of a fluid?

For horizontal flow, Bernoulli's principle gives $P + \frac{1}{2}\rho v^2 = \text{constant}$, so higher speed means lower pressure.

### 5.5. How is flow rate related to cross-sectional area and velocity?

Flow rate $Q = Av$. For an incompressible fluid, $A_1v_1 = A_2v_2$.

### 5.6. How can you study variation in velocity of a fluid in a hose of varying diameter?

Measure the volume collected per second and use $v = Q/A$ at different sections. Where the area is smaller, the speed is larger.

### 5.7. How does an object float or sink according to Archimedes' principle?

If the maximum upthrust is greater than or equal to the weight, the object floats; if the weight is greater than the maximum upthrust, it sinks.

### 5.8. How did Archimedes reportedly discover his principle?

He noticed that the water level rose when he entered a bath, realizing that the volume of displaced water equals the volume of the submerged body, leading to the idea of upthrust.

### 5.9. Why is standing near a fast moving train dangerous?

The fast air between the person and train has lower pressure. The higher pressure on the other side can push the person toward the train.

### 5.10. What are some potential applications of superfluidity?

Superfluids help study quantum fluids, frictionless flow, precision rotation sensors and low-temperature physics.

### 5.11. Differentiate between stress, strain and Young's modulus. Give SI units.

Stress = F/A, unit pascal. Strain = $\Delta L/L$, no unit. Young's modulus = stress/strain, unit pascal.

| Quantity | Definition | SI unit |
|---|---|---|
| Stress | Force per unit area | Pa |
| Strain | Fractional change in length | none |
| Young's modulus | Stress/strain | Pa |

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## Constructed Response Questions

**5.1 The ratio stress/strain is constant for small deformation. What happens for very large deformation?**

Beyond the proportional limit the ratio is no longer constant. The material may yield, deform plastically and finally fracture.

**5.2 Why does water spread on glass while mercury forms globules?**

For water, adhesion to glass is stronger than cohesion, so it spreads. For mercury, cohesion is stronger than adhesion, so it forms droplets.

**5.3 According to Bernoulli, pressure should be uniform in a pipe of uniform radius, but actually it decreases. Why?**

Real fluids have viscosity. Viscous forces dissipate energy as heat, so pressure falls along the pipe even when the radius is uniform.

**5.4 Why are aeroplane wings curved outward on top and flatter below?**

The shape makes air travel faster over the top, lowering pressure there. The higher pressure below produces lift.

**5.5 What is the difference between real fluid, ideal fluid and superfluid? Which really exists?**

An ideal fluid is incompressible and non-viscous, a model only. A real fluid has viscosity and exists in nature. A superfluid has zero viscosity below a critical temperature, such as liquid helium.

**5.6 Why is the study of superfluids important for low temperature physics?**

Superfluidity reveals quantum behaviour on a macroscopic scale, such as frictionless flow and quantized vortices, helping us understand matter near absolute zero.

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## Comprehensive / Long Questions

**5.1 Explain classification of solids with respect to atomic arrangements.**

Crystalline solids have a regular repeating lattice and sharp melting point. Amorphous solids have irregular arrangement and soften over a range. Polymeric solids consist of long chain molecules, natural or synthetic.

**5.2 What is Archimedes' principle? Explain it in detail for finding upthrust.**

A body immersed in a fluid experiences an upthrust equal to the weight of displaced fluid. For displaced volume V and fluid density ρ, upthrust $U = \rho g V$. It acts upward through the centre of buoyancy.

**5.3 Justify that mass remains conserved when a fluid flows through a pipe.**

For steady flow, mass entering per second equals mass leaving per second: $\rho A_1 v_1 = \rho A_2 v_2$. For an incompressible fluid this gives $A_1 v_1 = A_2 v_2$, the continuity equation.

### 5.4 Explain the term superfluidity.

Superfluidity is the state of a liquid with zero viscosity below a critical temperature. It can flow without friction, creep over container walls and sustain persistent vortices.

### 5.5 State and derive the equation of continuity.

For an incompressible fluid, volume flow rate is constant. In time $\Delta t$, volume entering $A_1v_1\Delta t$ equals volume leaving $A_2v_2\Delta t$, so $A_1v_1 = A_2v_2$.

### 5.6 State and prove Bernoulli's equation.

For steady, incompressible, non-viscous flow, work done by pressure forces equals change in kinetic and potential energy. This gives $P + \frac{1}{2}\rho v^2 + \rho gh = \text{constant}$ along a streamline.

### 5.7 Give some practical applications of Bernoulli's equation.

Aeroplane lift, swinging balls, filter pumps, carburettors, paint sprayers, Venturi meters and Torricelli's efflux theorem all use the relation between speed and pressure.

### 5.8 Define terminal velocity and show it is directly proportional to the square of radius.

Terminal velocity is the constant speed when drag plus upthrust equals weight. For a sphere, $6\pi\eta r v_t = (4/3)\pi r^3 \rho g$ (neglecting fluid density), giving $v_t = 2r^2\rho g/9\eta$, so $v_t \propto r^2$.

## Numerical Problems (fully worked)

### 5.1 A steel wire L = 2 m, A = 2 × 10⁻⁶ m², F = 400 N, Y = 2 × 10¹¹ N m⁻². Find extension.

**Given:** $L = 2$ m, $A = 2\times10^{-6}$ m², $F = 400$ N, $Y = 2\times10^{11}$ N m$^{-2}$

**Formula:** $\Delta L = FL/(AY)$

$\Delta L = 400\times2/(2\times10^{-6} \times 2\times10^{11}) = 0.002$ m.

**Answer: 0.002 m**

### 5.2 A spring k = 200 N m⁻¹ is stretched 0.5 m. Find elastic P.E.

**Given:** $k = 200$ N m$^{-1}$, $x = 0.5$ m

**Formula:** $E = \frac{1}{2}kx^2$

$E = \frac{1}{2} \times 200 \times 0.25 = 25$ J.

**Answer: 25 J**

### 5.3 Copper wire L = 3 m, A = 1 × 10⁻⁶ m², F = 500 N, Y = 1.1 × 10¹¹ N m⁻². Find stress and strain.

**Given:** $F = 500$ N, $A = 1\times10^{-6}$ m², $Y = 1.1\times10^{11}$ N m$^{-2}$

**Formula:** stress = F/A, strain = stress/Y

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Stress = $500/(1\times10^{-6}) = 5 \times 10^{8}$ N m$^{-2}$.

Strain = $5\times10^{8} / 1.1\times10^{11} \approx 0.00455$.

**Answer: $5 \times 10^{8}$ N m$^{-2}$, 0.00455**

### 5.4 A wood block mass 10 kg, density 600 kg m$^{-3}$ floats in water. Find buoyant force.

**Given:** m = 10 kg, floating

**Formula:** U = weight for floating body

U = mg = $10 \times 9.8 = 98$ N.

**Answer: 98 N**

### 5.5 Water flows in a pipe diameter 0.05 m at 2 m s$^{-1}$. Pipe narrows to 0.03 m. Find speed at narrow section.

**Given:** $d_1 = 0.05$ m, $v_1 = 2$ m s$^{-1}$, $d_2 = 0.03$ m

**Formula:** $A_1v_1 = A_2v_2$

$v_2 = v_1(d_1/d_2)^2 = 2 \times (0.05/0.03)^2 \approx 5.56$ m s$^{-1}$.

**Answer: 5.56 m s$^{-1}$**

### 5.6 Horizontal pipe: $v_1 = 3$ m s$^{-1}$, $P_1 = 200000$ Pa; at nozzle $P_2 = 101300$ Pa. Find exit speed. $\rho = 1000$ kg m$^{-3}$.

**Given:** $P_1 = 200000$ Pa, $P_2 = 101300$ Pa, $v_1 = 3$ m s$^{-1}$, $\rho = 1000$ kg m$^{-3}$

**Formula:** $P_1 + \frac{1}{2}\rho v_1^2 = P_2 + \frac{1}{2}\rho v_2^2$

$v_2 = \sqrt{[v_1^2 + 2(P_1 - P_2)/\rho]} = \sqrt{[9 + 197.4]} \approx 14.37$ m s$^{-1}$.

**Answer: 14.37 m s$^{-1}$**

### 5.7 A tank has a hole at depth 6 m. Find velocity of efflux.

**Given:** h = 6 m

**Formula:** $v = \sqrt{(2gh)}$

$v = \sqrt{(2\times9.8\times6)} \approx 10.8$ m s$^{-1}$. The book answer 9.9 m s$^{-1}$ corresponds to h ≈ 5 m.

**Answer: ≈ 10.8 m s$^{-1}$ (book: 9.9 m s$^{-1}$)**

Book answer appears to use a depth of about 5 m rather than 6 m.

### 5.8 Find terminal velocity of a raindrop radius 0.5 mm in air. $\eta = 1.9 \times 10^{-5}$ kg m$^{-1}$ s$^{-1}$, $\rho = 1000$ kg m$^{-3}$.

**Given:** $r = 5\times10^{-4}$ m, $\eta = 1.9\times10^{-5}$, $\rho = 1000$ kg m$^{-3}$

**Formula:** $v_t = 2r^2\rho g/9\eta$

$v_t = 2\times(5\times10^{-4})^2\times1000\times9.8/(9\times1.9\times10^{-5}) \approx 28.65$ m s$^{-1}$.

**Answer: 28.65 m s$^{-1}$**

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## Textbook checks flagged in this chapter

• Numerical 5.7: the printed depth 6 m gives about 10.8 m $s^{-1}$; the book answer 9.9 m $s^{-1}$ corresponds to about 5 m.

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